

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Show how comparisons can be used to determine if the following series converge or diverge.

a) $\sum_{n=1}^{\infty} \frac{n-1}{n^2 \sqrt{n}}$

Note that $\frac{n-1}{n^2 \sqrt{n}} \leq \frac{n}{n^2 \sqrt{n}} = \frac{n}{n^{5/2}} = \frac{1}{n^{3/2}}$ for all values of $n \geq 1$. The series $\sum_{n=1}^{\infty} \frac{n-1}{n^2 \sqrt{n}}$ and $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ contain only non-negative terms. Therefore, by the Comparison Test, since $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges (it is a p-series with $p = \frac{3}{2} > 1$) we know that $\sum_{n=1}^{\infty} \frac{n-1}{n^2 \sqrt{n}}$ must also converge.

b) $\sum_{n=1}^{\infty} \frac{1 + \sin(n)}{10^n}$

Note that $\frac{1 + \sin(n)}{10^n} \leq \frac{1+1}{10^n} = 2 \frac{1}{10^n} = 2 \left(\frac{1}{10} \right)^n$ for all values of $n \geq 1$. The series $\sum_{n=1}^{\infty} \frac{1 + \sin(n)}{10^n}$ and $\sum_{n=1}^{\infty} 2 \left(\frac{1}{10} \right)^n$ contain only non-negative terms. Therefore, by the Comparison Test, since $\sum_{n=1}^{\infty} 2 \left(\frac{1}{10} \right)^n$ converges (it is a geometric series with $r = \frac{1}{10}$) we know that $\sum_{n=1}^{\infty} \frac{1 + \sin(n)}{10^n}$ must also converge.

c) $\sum_{n=1}^{\infty} \frac{n+4^n}{n+6^n}$

Note that $\frac{n+4^n}{n+6^n} \leq \frac{n+4^n}{6^n} \leq \frac{4^n+4^n}{6^n} = \frac{2 \cdot 4^n}{6^n} = 2 \left(\frac{4}{6}\right)^n = 2 \left(\frac{2}{3}\right)^n$ for all values of $n \geq 1$. The series

$\sum_{n=1}^{\infty} \frac{n+4^n}{n+6^n}$ and $\sum_{n=1}^{\infty} 2 \left(\frac{2}{3}\right)^n$ contain only non-negative terms. Therefore, by the Comparison Test, since

$\sum_{n=1}^{\infty} 2 \left(\frac{2}{3}\right)^n$ converges (it is a geometric series with $r = \frac{2}{3}$) we know that $\sum_{n=1}^{\infty} \frac{n+4^n}{n+6^n}$ must also converge.

d) $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$

Note that $\frac{1}{n^2-1} > \frac{1}{n^2}$ for all values of $n \geq 2$. This isn't helpful for using the Comparison Test, but we

note that the series $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$ and $\sum_{n=2}^{\infty} \frac{1}{n^2}$ contain only positive terms. We calculate

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n^2}}{\frac{1}{n^2-1}} = \lim_{n \rightarrow \infty} \frac{n^2-1}{n^2} = \lim_{n \rightarrow \infty} \frac{n^2-1}{n^2} \cdot \frac{1/n^2}{1/n^2} = \lim_{n \rightarrow \infty} \frac{1-1/n^2}{1} = 1. \text{ Therefore, by the Limit Comparison Test,}$$

since 1 is both positive and finite both series have the same behavior. So, since $\sum_{n=2}^{\infty} \frac{1}{n^2}$ is a convergent

p-series (with $p = 2 > 1$) we know that $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$ must also converge.

e) $\sum_{n=6}^{\infty} \frac{\sqrt{n^3 + 2n + 1}}{3n^2 + 14n + 2}$

Trying to use direct comparisons here seems difficult because if we remove any terms from the denominator we make the expression larger, but if we remove any terms from the numerator we make the expression smaller. This creates complications with simplifying the expression and maintaining a consistent direction for our inequalities.

Instead, we consider the series $\sum_{n=6}^{\infty} \frac{\sqrt{n^3}}{n^2} = \sum_{n=6}^{\infty} \frac{1}{n^{1/2}}$ and note that this series and the given series contain only positive terms. We calculate that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\frac{\sqrt{n^3 + 2n + 1}}{3n^2 + 14n + 2}}{\frac{1}{n^{1/2}}} &= \lim_{n \rightarrow \infty} \frac{\sqrt{n^3 + 2n + 1}}{3n^2 + 14n + 2} \cdot n^{1/2} = \lim_{n \rightarrow \infty} \frac{\sqrt{n^4 + 2n^2 + n}}{3n^2 + 14n + 2} \\ &= \lim_{n \rightarrow \infty} \frac{\sqrt{n^4 + 2n^2 + n}}{3n^2 + 14n + 2} \cdot \frac{1/n^2}{1/n^2} = \lim_{n \rightarrow \infty} \frac{\sqrt{1 + \frac{2}{n^2} + \frac{1}{n^3}}}{3 + \frac{14}{n} + \frac{2}{n^2}} = \lim_{n \rightarrow \infty} \frac{\sqrt{1}}{3} \\ &= \frac{1}{3} \end{aligned}$$

Therefore, by the Limit Comparison Test, since $\frac{1}{3}$ is both positive and finite both series have the same

behavior. So, since $\sum_{n=2}^{\infty} \frac{1}{n^{1/2}}$ is a divergent p-series (with $p = \frac{1}{2} < 1$) we know that $\sum_{n=6}^{\infty} \frac{\sqrt{n^3 + 2n + 1}}{3n^2 + 14n + 2}$ must also diverge.

f) $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

Note that $\frac{n!}{n^n} = \frac{n(n-1)(n-2)\dots 3 \cdot 2 \cdot 1}{n \cdot n \cdot \dots \cdot n} = \frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdot \dots \cdot \frac{3}{n} \cdot \frac{2}{n} \cdot \frac{1}{n} \leq \frac{2}{n} \cdot \frac{1}{n} = \frac{2}{n^2}$ for all values of $n \geq 1$.

The series $\sum_{n=1}^{\infty} \frac{n + 4^n}{n + 6^n}$ and $\sum_{n=1}^{\infty} 2\left(\frac{2}{3}\right)^n$ contain only non-negative terms. Therefore, by the Comparison

Test, since $\sum_{n=1}^{\infty} 2\left(\frac{2}{3}\right)^n$ converges (it is a geometric series with $r = \frac{2}{3}$) we know that $\sum_{n=1}^{\infty} \frac{n + 4^n}{n + 6^n}$ must also converge.